

● MEDIUM

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

04

Q1

ADVANCED MATH

On April 1, there were 233 views of an advertisement posted on a website. Every 2 days after April 1, the number of views of the advertisement had increased by 70% of the number of views 2 days earlier. The function f gives the predicted number of views x days after April 1. Which equation defines f ?

- A. $f(x) = 2330.70^{\frac{x}{2}}$
- B. $f(x) = 2330.70^{2x}$
- C. $f(x) = 2331.70^{\frac{x}{2}}$
- D. $f(x) = 2331.70^{2x}$

$$f(x) = (x + 6)(x + 5)(x - 4)$$

The function f is given. Which table of values represents $y = f(x) - 3$?

A.

x	y
-6	-9
-5	-8
4	1

B.

x	y
-6	-3
-5	-3
4	-3

C.

x	y
-6	-3
-5	-2
4	7

D.

x	y
-6	3
-5	3
4	3

$$N(d) = 115(0.90)^d$$

The function N defined above can be used to model the number of species of brachiopods at various ocean depths d , where d is in hundreds of meters. Which of the following does the model predict?

- A. For every increase in depth by 1 meter, the number of brachiopod species decreases by 115.
- B. For every increase in depth by 1 meter, the number of brachiopod species decreases by 10%.
- C. For every increase in depth by 100 meters, the number of brachiopod species decreases by 115.
- D. For every increase in depth by 100 meters, the number of brachiopod species decreases by 10%.

$$y = 0.25x^2 - 7.5x + 90.25$$

The equation gives the estimated stock price y , in dollars, for a certain company x days after a new product launched, where $0 \leq x \leq 20$. Which statement is the best interpretation of $x, y = 1, 83$ in this context?

- A. The company's estimated stock price increased \$ 83 every day after the new product launched.
- B. The company's estimated stock price increased \$ 1 every 83 days after the new product launched.
- C. 1 day after the new product launched, the company's estimated stock price is \$ 83 .
- D. 83 days after the new product launched, the company's estimated stock price is \$ 1.

A scientist initially measures 12,000 bacteria in a growth medium. 4 hours later, the scientist measures 24,000 bacteria. Assuming exponential growth, the formula $P = C2^{rt}$ gives the number of bacteria in the growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. What is the value of r ?

- A. $\frac{1}{12,000}$
- B. $\frac{1}{4}$
- C. 4
- D. 12,000

For the exponential function f , the value of $f(0)$ is c , where c is a constant. Of the following equations that define the function f , which equation shows the value of c as the coefficient or the base?

- A. $f(x) = 221.5^{x+1}$
- B. $f(x) = 331.5^x$
- C. $f(x) = 49.51.5^{x-1}$
- D. $f(x) = 74.251.5^{x-2}$

Q7

ADVANCED MATH

$$f(x) = x^3 + 3x^2 - 6x - 1$$

For the function f defined above, what is the value of $f(-1)$?

- A. -11
- B. -7
- C. 7
- D. 11

Q8

ADVANCED MATH

A function p estimates that there were 2,000 animals in a population in 1998. Each year from 1998 to 2010, the function estimates that the number of animals in this population increased by 3% of the number of animals in the population the previous year. Which equation defines this function, where px is the estimated number of animals in the population x years after 1998?

- A. $px = 2,0003^x$
- B. $px = 2,0001.97^x$
- C. $px = 2,0001.03^x$
- D. $px = 2,0000.97^x$

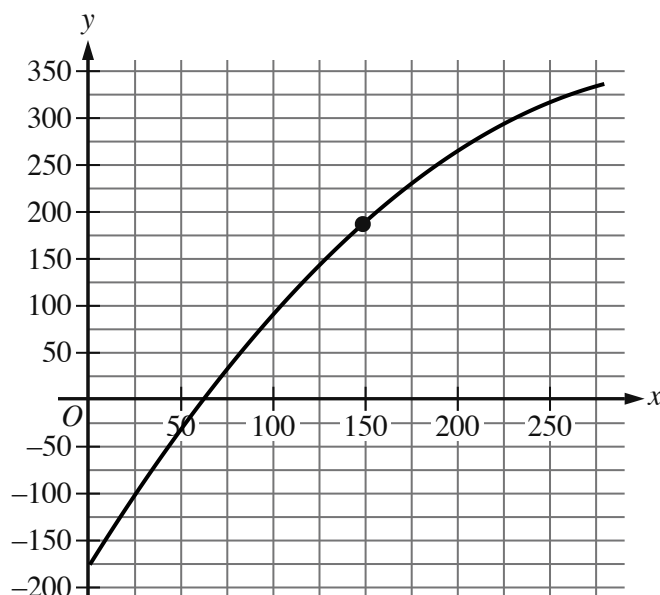
The number of bacteria in a liquid medium doubles every day. There are 44,000 bacteria in the liquid medium at the start of an observation. Which represents the number of bacteria, y , in the liquid medium t days after the start of the observation?

A. $y = \frac{1}{2}44,000^t$

B. $y = 244,000^t$

C. $y = 44,000\frac{1}{2}^t$

D. $y = 44,0002^t$



- The parabola opens downward.
- The parabola passes through the following approximate points:
 - (50 comma negative 33.83)
 - (149.02 comma 186.05)
 - (250 comma 315.37)

The graph shows the estimated boiling point y , in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point 149.02186.05?

- A. A normal paraffin with a molecular weight of 186.05 grams per mole has an estimated boiling point of 149.02 degrees Celsius.
- B. A normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.
- C. The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.

- D. The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of 149.02 grams per mole and an estimated boiling point of 186.05 degrees Celsius.